Neil Panchal

Clarksburg, MD | neilpanchal08@gmail.com | linkedin.com/in/neilpanch

Education

University of Maryland, College Park - Fischell Department of Bioengineering — **College Park, MD**
Bachelor of Science in Bioengineering | Virtus Living-Learning Community — *Expected May 2030*
Poolesville High School - Global Ecology Magnet Program — **Poolesville, MD**
GPA: 3.98/4.00 (Unweighted) — *June 2026*

Experience

Biophysics Undergraduate Researcher — **September 2026 – Present**
University of Maryland - Fischell Department of Bioengineering (Aranda-Espinoza Lab) — *College Park, MD*
- Model how adhesion, confinement, and shear trigger neutrophil extracellular trap (NET) formation, toward one unified model
- Reviewed ~9,700 NET papers and found only 2.5% address mechanics and 0.3% use quantitative models; ranked 11 open gaps and 8 experiments the lab can run with existing equipment
- Proposed a framework linking all three routes through calcium/calpain-driven nuclear envelope failure; scoping a review paper
- Next, will test NET physics experimentally, starting with a substrate stiffness dose-response no published study has fit

Biomedical Imaging Research Intern — **June 2026 – Present**
University of Maryland - Fischell Department of Bioengineering (Gan Lab) — *College Park, MD*
- Quantify intraoperative bone viability from white-light and ICG fluorescence imaging (Python, OpenCV, pydicom) with Dartmouth collaborators, replacing subjective visual inspection
- Segment viable vs. non-viable bone at 0.89 Dice by fusing two fluorescence views, with a calibrated confidence measure that flags unreliable cases (calibration error 0.10 to 0.06); localize non-perfused bone at 0.85 patient-level AUC ($p = 0.0001$)
- Scaled data from 50 to 270 cases (207 patients), raising held-out ICG scores from 0.878 to 0.906; added nnU-Net and UNet++
- Drafting an ISBI 2027 paper; poster at the Mickey Dale Engineering the Future of Organ Transplantation Symposium (Oct 2026)

Undergraduate Research Intern — **June 2026 – Present**
George Mason University - Vaisman Lab — *Fairfax, VA (Remote)*
- Generate frozen ESM-2 (650M) protein embeddings as input to a path-based graph neural network (GNN) for knowledge-graph link prediction (Protein2GO) on a 160K-node biomedical graph

Youth Conservation Crew Member (Paid Internship) — **June 2026 – July 2026**
MobilizeGreen — *Remote*
- Co-developed an interactive drought and wildfire simulator (Hugging Face) comparing flame spread under set drought and wind
- Wrote an independent article on development threats to Ten Mile Creek, which feeds the backup water supply for ~4.3M people

Lab Researcher (Paid) — **December 2025 – February 2026**
University of Maryland - Institute for Bioscience & Biotechnology Research — *Rockville, MD*
- Ran the first ablation study on negative sampling for protein language model (ESM-2) training in antimicrobial peptide (AMP) discovery; DisProt-based negatives outperformed all alternatives ($F1 = 0.981$ vs. 0.931–0.975)
- Computationally designed and independently validated NeKtar-1 (AMP_031) under BSL-2 (bactericidal, hemolysis assays), killing 99.9–99.999% of drug-resistant *E. coli* and *P. aeruginosa*; published in the *National High School Journal of Science*

Research Intern — **May 2025 – January 2026**
Johns Hopkins University - Biomedical Engineering — *Baltimore, MD*
- Cultured patient-derived cancer organoids and ran the first benchmark of 10 focus metrics for extended depth of field (EDF) reconstruction across 990 images; metric choice explained 32% of output-quality variance ($p < 0.001$)
- Built two open-source Python/Streamlit tools, including OrganoidEDF, which outputs a 16-bit all-in-focus image and depth map

Engineering Research Intern — **July 2025 – August 2025**
ESTEEM/SER-Quest Internship - Clark School of Engineering, UMD — *College Park, MD*
- Selected as 1 of 15 from 125+ applicants; co-developed a bias-aware AI advising chatbot (ML + regex), published on arXiv

Skills

Laboratory: BSL-2 containment, aseptic technique, cell and organoid culture, brightfield/z-stack microscopy, bactericidal assays (serial dilutions, CFU plating, MIC/MBC determination), hemolysis assays (plate reader/spectrophotometry)
Computational: Python (ML/AI), PyTorch, PyTorch Geometric, Hugging Face, ESM-2 fine-tuning, GNNs, image segmentation (nnU-Net, UNet++), OpenCV, pydicom, scikit-learn, pandas, NumPy, Streamlit, LaTeX/Overleaf, AutoCAD, Onshape
Research: experimental design, systematic literature review, ablation studies, benchmarking, model calibration, statistical analysis (ROC/AUC, dose-response), train/test leakage auditing, protein language models, manuscript and poster writing

Awards & Honors

Regeneron Biomedical Science Award | AstraZeneca Exceptional Project Award | HOSA 2nd Place State Biomedical Debate
MRIGlobal 1st Place Project Award | Science Montgomery 3rd Place | Metro Bethesda Rotary James Phang Aspiring Scientist Award